Lecture Scribe Exam Reference

L20.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing
to consult any other material.

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presented in the slides. The reasoning must be explicit but strictly limited to what
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2. When the lecture introduces properties, axioms, rules, corollaries, propositions,
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   accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
   written on the slides. Do NOT fill in missing steps, add justifications, or extend
   arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
   full detail:
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- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major concept must be written out completely, with the same level of detail as in the lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow, ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory, or generalizations. If a proof, derivation, or explanation is omitted or only partially shown in the slides, present it only to that same extent.

Below is a **complete exam-oriented lecture scribe** based strictly on the uploaded slides . The structure, wording, derivations, and examples follow the lecture exactly, without adding external information.

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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1. Random Vector Representation

1.1 Vector Form of Multiple Random Variables

Definition: Random Vector

For random variables $X_1, X_2, \ldots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Explanation (from slide)

- X is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

---

2. Mean Vector

2.1 Expectation of a Random Vector

Definition

The mean vector of X is:

$$\mu_X = E[X]$$

Explanation

- Represents the **average value of the random vector**.
- Each element corresponds to the **mean of one random variable**.

2.2 Structure of Mean Vector

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

---

3. Correlation Matrix

3.1 Correlation Between Multiple Random Variables

Definition

$$R_{XX} = E[XX^T]$$

Element-wise:

$$R_{XX}(i,j) = E[X_i X_j]$$

Condition

$$i, j \leq N$$

3.2 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

---

4. Covariance Matrix

4.1 Mathematical Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

4.2 Matrix Form (Practical View)

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

4.3 Property: Symmetry

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

- The matrix is a **mirror of itself**.

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5. Linear Transformation of Random Vectors

5.1 Linear Model

$$Y = AX + b$$

Components

- A: Transformation matrix (scales and rotates data)
- b: Constant vector (shifts/translates center)
- X: Input random vector

5.2 Expanded Form (N-dimensional)

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2$$

$$\vdots$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$$

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6. Mean of Transformed Vector

6.1 Definition

$$\mu_Y = E[Y]$$

6.2 Derivation

Substitute $Y = AX + b$:

$$\mu_Y = E[AX + b]$$

Apply linearity:

$$\mu_Y = E[AX] + E[b]$$

6.3 Final Result

$$\mu_Y = A\mu_X + b$$

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7. Correlation Matrix Transformation

7.1 Definition

$$R_{YY} = E[YY^T] = E[(AX+b)(AX+b)^T]$$

7.2 Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

7.3 Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

7.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

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8. Covariance Matrix Transformation

8.1 Key Observation

$$Y - \mu_Y = (AX+b) - (A\mu_X + b) = A(X - \mu_X)$$

- The deviation from mean is **independent of shift** b.

8.2 Derivation

$$C_{YY} = E[(Y-\mu_Y)(Y-\mu_Y)^T]$$

$$C_{YY} = E[A(X-\mu_X)(X-\mu_X)^TA^T]$$

8.3 Simplified Result

$$C_{YY} = AC_{XX}A^T$$

Note

- Shift b does **not affect covariance**.

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9. Summary of Linear Transformation

If $Y = AX + b$:

Property	Result
Mean	$\mu_Y = A\mu_X + b$
Covariance	$C_{YY} = AC_{XX}A^T$
Correlation	$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

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10. Example: Numerical Linear Transformation

10.1 Given

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

10.2 (a) Mean Vector

Using:

$$\mu_Y = A\mu_X$$

Multiplication steps (as shown on slide):

$$\mu_Y = \begin{bmatrix} (1)(2) + (0)(-1) + (-1)(1) \\ (0)(2) + (-1)(-1) + (1)(1) \\ (-1)(2) + (1)(-1) + (0)(1) \end{bmatrix}$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

10.3 (b) Correlation Matrix

Using:

$$R_{YY} = AR_{XX}A^T$$

Step 1: Compute AR_{XX}

$$= \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

Step 2: Multiply by A^T

Final Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

10.4 (c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Outer Product

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Final Subtraction

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

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11. Gaussian Distributions

11.1 Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

11.2 Joint Gaussian (2D)

$$f_{XY}(x,y) = \frac{\exp\left(-\frac{1}{2(1-\rho_{XY}^2)}[\cdots]\right)}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}}$$

(Full expression shown exactly in slides)

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12. Multivariate Gaussian Distribution

12.1 Definition

$$X = [X_1, X_2, \ldots, X_N]^T$$

12.2 General PDF

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

12.3 Components

- μ_X: Mean vector
- C_{XX}: Covariance matrix

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13. Example: 2D Joint Gaussian PDF

Step 1: Determinant

$$\det(C_{XX}) = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$$

Step 2: Inverse

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})} \mathrm{adj}(C_{XX})$$

Expanded form shown in slides.

Step 3: Quadratic Form

$$Q(x) = (x - \mu_X)^T C_{XX}^{-1} (x - \mu_X)$$

Final Expression

$$Q(x) = \frac{1}{1 - \rho^2} \left[\left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \frac{x_2 - \mu_2}{\sigma_2} \right) + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right]$$

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14. Uncorrelated Gaussian Variables

14.1 Statement

If variables are **mutually uncorrelated**:

$$\mathrm{Cov}(X_i, X_j) = 0, \quad i \neq j$$

14.2 Result

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

14.3 Interpretation

- Covariance matrix is **diagonal**
- Joint PDF **factorizes into individual PDFs**

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15. Homework

Problem

$$\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\mathrm{Cov}(X,Y)$$

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16. Case Study: Smart City Traffic + Weather System

16.1 Random Vector Definition

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where:

- X_1: Traffic flow
- X_2: Speed
- X_3: Rainfall

- X_4: Temperature
- X_5: AQI

16.2 Mean Vector Example

$$\mu = E[X]$$

Example values:

- Traffic = 120
- Speed = 35
- Rainfall = 2
- Temp = 32
- AQI = 180

16.3 Covariance Interpretation

- More rain ⇒ traffic increases
- More rain ⇒ speed decreases
- More traffic ⇒ pollution increases
- Higher temperature ⇒ pollution disperses

Covariance matrix = "**dependency map of the city**"

16.4 Joint Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

16.5 Linear Transformation (Congestion Index)

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

Also written as:

$$Y = a^T X$$

16.6 Inference

If rainfall increases:

- Speed ↓
- Traffic ↑
- Pollution ↑

End of Lecture

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Lecture 20: Linear Transformations and Expectations of Multiple RVs

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1. Statistical Representation of Random Vectors

1.1 Vector Form of Multiple Random Variables

Random Vector

For random variables $X_1, X_2, \ldots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Statements (as given)

- X is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

---

2. Mean Vector, Correlation, and Covariance Matrices

2.1 Mean Vector

Expectation of a Random Vector

Mean Vector Definition

$$\mu_X = E[X]$$

Statements

- Represents the **average value of the random vector**.
- Each element corresponds to the **mean of one random variable**.

2.2 Example Structure of Mean Vector

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

2.3 Correlation Matrix

Correlation Between Multiple RVs

Correlation Matrix Definition

$$R_{XX} = E[XX^T]$$

where i and $j \leq N$,

$$R_{XX}(i,j) = E[X_i X_j]$$

2.4 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

2.5 Covariance Matrix

Mathematical Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

2.6 Practical Matrix Form

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

2.7 Symmetry Property

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

- The matrix is a "**mirror**" **of itself**.

---

3. Linear Transformations

3.1 The Linear Model

Consider the transformation:

$$Y = AX + b$$

Definitions (as given)

- A: Transformation matrix (scales and rotates the data)
- b: Constant vector (shifts/translates the center)
- X: Input random vector

3.2 N-Dimensional Expansion

For N-dimensional vectors:

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2$$

$$\vdots$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$$

---

4. Mapping Properties under $Y = AX + b$

4.1 Mean of Transformed Vector

Definition

$$\mu_Y = E[Y]$$

4.2 Derivation (Step-by-Step)

- Substitute $Y = AX + b$ into expectation:

$$\mu_Y = E[AX + b]$$

- Apply linearity:

$$\mu_Y = E[AX] + E[b]$$

4.3 Result

$$\mu_Y = A\mu_X + b$$

4.4 Correlation Matrix Transformation

Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

Expansion of Terms

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

4.5 Covariance Matrix Transformation

Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$

- The deviation from the mean is **independent of the shift** b.

Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

Simplified Result

$$C_{YY} = AC_{XX}A^T$$

Note

- The shift b does not affect the covariance (spread).

4.6 Summary Table

If $Y = AX + b$:

- Mean:

$$\mu_Y = A\mu_X + b$$

- Covariance:

$$C_{YY} = AC_{XX}A^T$$

- Correlation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

---

5. Example: Numerical Linear Transformation

5.1 Given

$$Y = AX$$

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

5.2 (a) Mean Vector of Y

Using:

$$\mu_Y = A\mu_X$$

Computation

$$\mu_Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(2) + (0)(-1) + (-1)(1) \\ (0)(2) + (-1)(-1) + (1)(1) \\ (-1)(2) + (1)(-1) + (0)(1) \end{bmatrix}$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

5.3 (b) Correlation Matrix of Y

Using:

$$R_{YY} = AR_{XX}A^T$$

Step 1: Compute AR_{XX}

$$= \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

Step 2: Multiply by A^T

$$\begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

5.4 (c) Covariance Matrix of Y

Using:

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Outer Product

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} [1\ 2\ -3] = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Final Subtraction

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

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6. Multivariate Gaussian Distributions

6.1 Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

6.2 Joint Gaussian (2D)

For two RVs X and Y, the joint PDF is:

(full expression exactly as shown in slide)

6.3 Multivariate Gaussian Distribution

For an N-dimensional random vector:

$$X = [X_1, X_2, \ldots, X_N]^T$$

6.4 General Vector PDF

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

6.5 Structure Components

- μ_X: Mean vector ($N \times 1$)
- C_{XX}: Covariance matrix ($N \times N$)

---

7. Example: 2D Joint Gaussian PDF

7.1 Given

$$\mu_X = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

7.2 Determinant

$$\det(C_{XX}) = \sigma_1^2\sigma_2^2(1-\rho^2)$$

7.3 Inverse

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})}\mathrm{adj}(C_{XX})$$

7.4 Quadratic Form

$$Q(x) = (x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)$$

7.5 Final Expression

$$Q(x) = \frac{1}{1-\rho^2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]$$

---

8. Special Case: Uncorrelated Gaussian Variables

8.1 Key Insight

For mutually uncorrelated Gaussian random variables:

$$\mathrm{Cov}(X_i, X_j) = 0, \quad \forall i \neq j$$

8.2 Covariance Matrix Form

Diagonal matrix:

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots \\ 0 & \sigma_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

8.3 Resulting PDF

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2}\right)$$

8.4 Interpretation

- Covariance matrix is diagonal
- Joint Gaussian PDF factorizes into product of individual PDFs

---

9. Homework

$$\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\mathrm{Cov}(X,Y)$$

---

10. Case Study: Smart City Traffic + Weather System

10.1 Step 1: Define N Random Variables

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where:

- X_1: Traffic flow (vehicles/min)
- X_2: Average vehicle speed (km/h)
- X_3: Rainfall intensity (mm/hr)
- X_4: Temperature (°C)
- X_5: Air pollution index (AQI)

10.2 Step 2: Mean Vector

$$\mu = E[X]$$

Example values:

- Avg traffic flow = 120
- Avg speed = 35
- Avg rainfall = 2
- Avg temp = 32
- Avg AQI = 180

10.3 Step 3: Covariance Matrix

Covariance matrix = "**dependency map of the city**"

Examples:

- $\mathrm{Cov}(X_1, X_3) > 0$
- $\mathrm{Cov}(X_2, X_3) < 0$
- $\mathrm{Cov}(X_1, X_5) > 0$
- $\mathrm{Cov}(X_4, X_5) < 0$ (maybe)

10.4 Step 4: Joint Gaussian Distribution

$$X \sim \mathcal{N}(\mu, \Sigma)$$

This means:

- Mean → typical conditions
- Covariance → interdependencies

10.5 Step 6: Transformation

Congestion index:

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

$$Y = a^T X$$

10.6 Inference

If rainfall increases suddenly:

- Speed ↓
- Traffic ↑
- Pollution ↑

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subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules, corollaries, propositions,
   or results, present each one separately, with its exact mathematical form and
   accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
written on the slides. Do NOT fill in missing steps, add justifications, or extend
arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
full detail:
- define all quantities and events exactly as done in the lecture,
- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

Below is a **strictly slide-faithful, exam-oriented lecture scribe** constructed only from the provided lecture slides .

All content follows the exact lecture order, with definitions, derivations, and examples reproduced step by step exactly as shown.

Lecture 20: Linear Transformations and Expectations of Multiple RVs

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1. Outline of the Lecture

The lecture covers:

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

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2. Random Vector Representation

2.1 Vector Form of Multiple Random Variables

Definition: Random Vector

For random variables $X_1, X_2, \ldots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Statements

- X is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

---

3. Mean Vector

3.1 Expectation of a Random Vector

Mean Vector Definition

$$\mu_X = E[X]$$

Statements

- Represents the **average value of the random vector**.
- Each element corresponds to the **mean of one random variable**.

3.2 Example Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

---

4. Correlation Matrix

4.1 Correlation Between Multiple RVs

Correlation Matrix Definition

$$R_{XX} = E[XX^T]$$

with:

$$R_{XX}(i,j) = E[X_i X_j], \quad i,j \leq N$$

4.2 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

---

5. Covariance Matrix

5.1 Mathematical Definition

For a random vector X, the covariance matrix is:

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

5.2 Practical Structure

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

5.3 Symmetry Property

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

- The matrix is a "**mirror" of itself**.

---

6. Linear Transformation of Random Vectors

6.1 The Linear Model

$$Y = AX + b$$

Definitions

- A: Transformation matrix (scales and rotates the data)
- b: Constant vector (shifts/translates the center)
- X: Input random vector

6.2 Expanded Form

For N-dimensional vectors:

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2$$

$$\vdots$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$$

---

7. Mean of Transformed Vector

7.1 Definition

$$\mu_Y = E[Y]$$

7.2 Derivation (exact steps)

- Substitute:

$$\mu_Y = E[AX + b]$$

- Apply linearity:

$$\mu_Y = E[AX] + E[b]$$

7.3 Result

$$\mu_Y = A\mu_X + b$$

---

8. Correlation Matrix Transformation

8.1 Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

8.2 Expansion of Terms

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

8.3 Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

8.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

---

9. Covariance Matrix Transformation

9.1 Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$

- The deviation from the mean is **independent of the shift** b.

9.2 Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

9.3 Simplified Result

$$C_{YY} = AC_{XX}A^T$$

Note

- The shift b does not affect the covariance (spread).

---

10. Summary: Effects of Linear Transformation

If $Y = AX + b$:

- Mean:

$$\mu_Y = A\mu_X + b$$

- Covariance:

$$C_{YY} = AC_{XX}A^T$$

- Correlation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

---

11. Example: Numerical Linear Transformation

11.1 Given

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

11.2 (a) Mean Vector

Using:

$$\mu_Y = A\mu_X$$

Step-by-step multiplication

$$\mu_Y = \begin{bmatrix} (1)(2) + (0)(-1) + (-1)(1) \\ (0)(2) + (-1)(-1) + (1)(1) \\ (-1)(2) + (1)(-1) + (0)(1) \end{bmatrix}$$

Result

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

11.3 (b) Correlation Matrix

Using:

$$R_{YY} = AR_{XX}A^T$$

Step 1: Compute AR_{XX}

$$= \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

Step 2: Multiply by A^T

$$\begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

11.4 (c) Covariance Matrix

Using:

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Outer Product

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Final Subtraction

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

12. Gaussian Distributions

12.1 Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

12.2 Joint Gaussian (2D)

For two RVs X and Y, the joint PDF is given (full expression shown in slide).

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13. Multivariate Gaussian Distribution

13.1 Definition

$$X = [X_1, X_2, \ldots, X_N]^T$$

13.2 General Vector PDF

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

13.3 Structure Components

- μ_X: Mean vector ($N \times 1$)
- C_{XX}: Covariance matrix ($N \times N$)

---

14. Example: 2D Joint Gaussian PDF

14.1 Given

$$\mu_X = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

14.2 Determinant

$$\det(C_{XX}) = \sigma_1^2\sigma_2^2(1-\rho^2)$$

14.3 Inverse

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})}\mathrm{adj}(C_{XX})$$

14.4 Quadratic Form

$$Q(x) = (x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)$$

14.5 Final Expression

$$Q(x) = \frac{1}{1-\rho^2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]$$

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15. Special Case: Uncorrelated Gaussian Variables

15.1 Key Insight

$$\mathrm{Cov}(X_i, X_j) = 0, \quad \forall i \neq j$$

15.2 Covariance Matrix Simplification

Diagonal matrix form.

15.3 Result

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

15.4 Interpretation

- Covariance matrix is diagonal
- Joint Gaussian PDF factorizes into product of individual PDFs

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16. Homework

$$\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\mathrm{Cov}(X, Y)$$

---

17. Case Study: Smart City Traffic + Weather System

17.1 Step 1: Define Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where:

- X_1: Traffic flow
- X_2: Speed
- X_3: Rainfall
- X_4: Temperature
- X_5: AQI

17.2 Step 2: Mean Vector

$$\mu = E[X]$$

Example values:

- Traffic = 120
- Speed = 35
- Rainfall = 2
- Temperature = 32
- AQI = 180

17.3 Step 3: Covariance Matrix

Examples:

- $\mathrm{Cov}(X_1, X_3) > 0$
- $\mathrm{Cov}(X_2, X_3) < 0$

- $\mathrm{Cov}(X_1, X_5) > 0$
- $\mathrm{Cov}(X_4, X_5) < 0$ (maybe)

Covariance matrix = "**dependency map of the city**"

17.4 Step 4: Joint Gaussian Distribution

$$X \sim \mathcal{N}(\mu, \Sigma)$$

17.5 Step 6: Transformation

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

$$Y = a^T X$$

17.6 Inference

If rainfall increases suddenly:

- Speed ↓
- Traffic ↑
- Pollution ↑

End of Lecture 20